

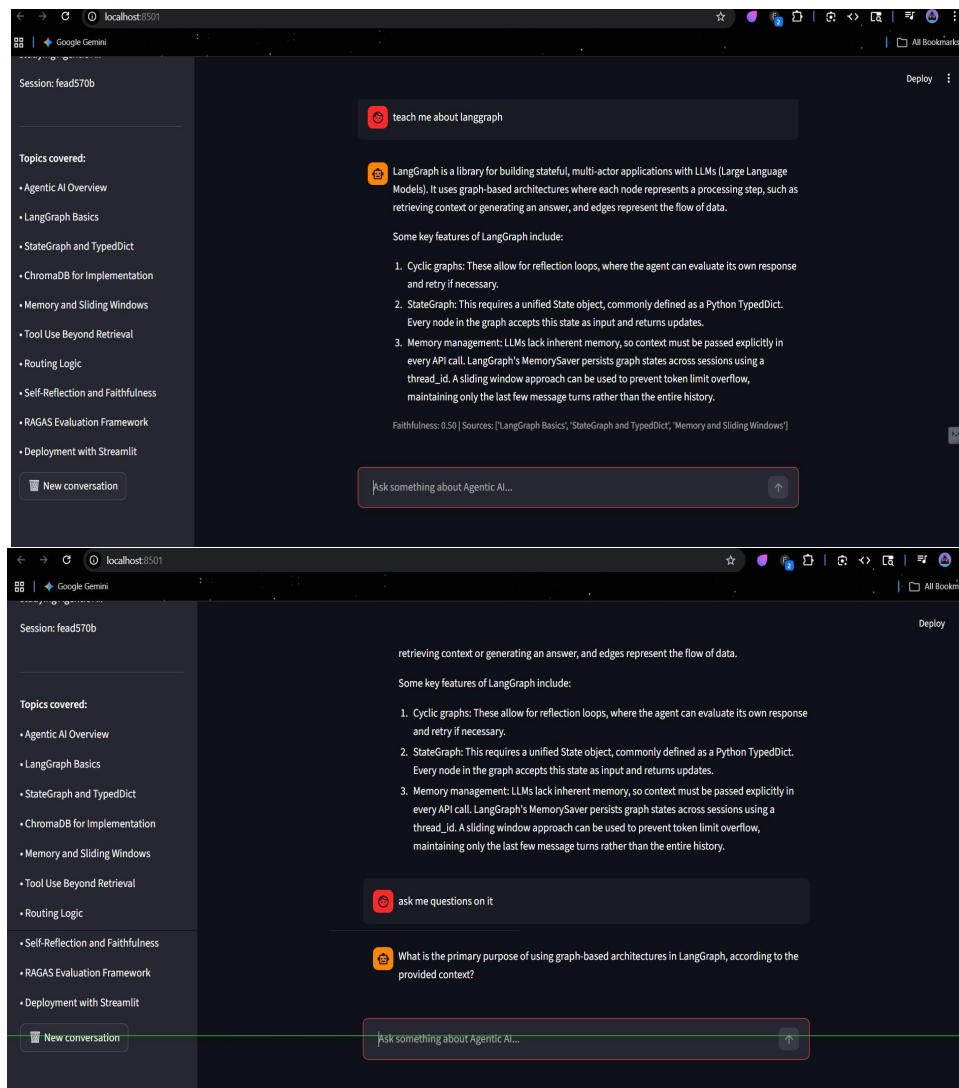
Agentic AI Course Assistant

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1. Project Overview

The Agentic Course Assistant is a Capstone Project designed to help 4th-year B.Tech students intelligently navigate their Agentic AI coursework. It acts as an autonomous tutor, utilizing a state-of-the-art intelligent agent architecture that goes beyond simple text generation to feature sophisticated routing, retrieval, evaluation, and tool-use capabilities.



2. System Architecture

The project framework consists of two core components that establish a robust backend and an interactive frontend:

- `agent.py`: The core 'brain' of the AI containing LangGraph states, Retrieval-Augmented Generation (RAG) logic, and model interaction logic.
- `capstone_streamlit.py`: The interactive user interface built dynamically with Streamlit.

3. Core Agent Design (Backend)

The backend agent is orchestrated using LangGraph to create a cyclic, stateful flowchart (StateGraph). The agent determines the optimal course of action dynamically as questions arrive. Key elements of this architecture include:

- **Knowledge Base Integration**: A predefined domain expertise base containing essential syllabus topics like System Routing Logic, ChromaDB, RAGAS Evaluation, and LangGraph Basics.
- **Vector Database (ChromaDB)**: The application encodes text snippets with 'SentenceTransformer' ('all-MiniLM-L6-v2') into a localized ChromaDB instance. When a student poses a query, the system retrieves the most semantically relevant chunks of information to use as context.
- **Large Language Model**: Powered by Groq's high-speed inference engine using the LLaMA 3.3 (70B parameter) model for highly capable semantic reasoning.
- **Intelligent Router**: Upon receiving a question, the routing node critically evaluates whether the agent should query the knowledge base ('retrieve' route), rely purely on conversation history ('memory_only' route), or execute external actions ('tool' route, such as retrieving the current date/time). This minimizes unnecessary database queries and reduces latency.
- **Evaluation & Self-Reflection Loop**: The generated answer undergoes a 'faithfulness' check by the `eval_node`. If the model hallucinated information not present in the retrieved texts (dropping below a 0.7 precision metric), the StateGraph forces the LLM to discard its answer and retry (up to 2 times), strictly adhering to factual data.
- **Long-Term Memory Management**: Utilizing LangGraph's 'MemorySaver', the agent accurately tracks conversational states using a `thread_id`, preventing token limit overflow by retaining historical context seamlessly.

4. Web Interface (Frontend)

The Streamlit UI provides a seamless user interaction layer directly tied to the AI infrastructure:

- **Session Flow Management**: Every unique web-tab instance or initialized 'New conversation' button correctly assigns a fresh UUID `thread_id`. This isolates conversational states, preventing test context leakage between consecutive sessions.

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- The screenshot shows a Visual Studio Code interface. On the left, the Explorer sidebar displays a project structure with files like __pycache__, .env, agent.py, Agentic AI Project Helper Document.doc..., capstone_streamlit.py, day13_capstone.ipynb, and update_notebook.py. The main editor area shows the code in agent.py, which defines a class AgentBuilder with methods eval_node and save_node. The terminal at the bottom shows the command to run the Streamlit app, followed by local and network URLs. It also displays a warning about unauthenticated requests to the Hugging Face Hub and a loading report for the BertModel. A note explains that UNEXPECTED values can be ignored when loading from different tasks or architectures.

The Agentic Course Assistant codebase is fully completed and smoothly operational. Crucial dependencies encompassing LangChain, langgraph, Groq, ChromaDB, and Streamlit have been integrated successfully. The application server runs efficiently locally, resulting in highly responsive and robust academic guidance to assist students with their coursework.